

7.1 Scaling Laws

Loss falls as a power of compute, data, and parameters — until you starve one of them.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Bigger models, trained on more data with more compute, tend to get better at next-token prediction.

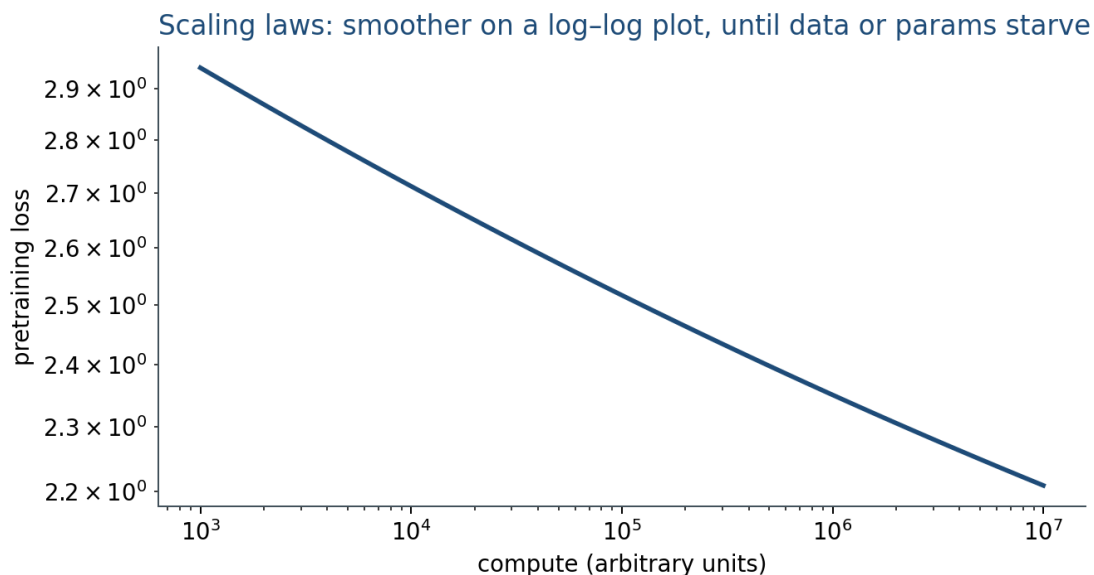
Scaling laws make that trend a curve you can fit, not a slogan. They do not promise a better chatbot; they promise a lower **pretraining loss** if the recipe stays the same.

1. Kaplan: loss as a power of compute

Kaplan et al. (2020) fit language-model cross-entropy against compute C (and against parameters N , dataset size D) and saw smooth **power laws** over many orders of magnitude:

$$L(C) \approx aC^{-b} + L_{\infty}.$$

The irreducible term L_{∞} is entropy of the data plus whatever the architecture cannot represent. The interesting bit is the slope $-b$: extra FLOPs still buy loss, with diminishing returns.

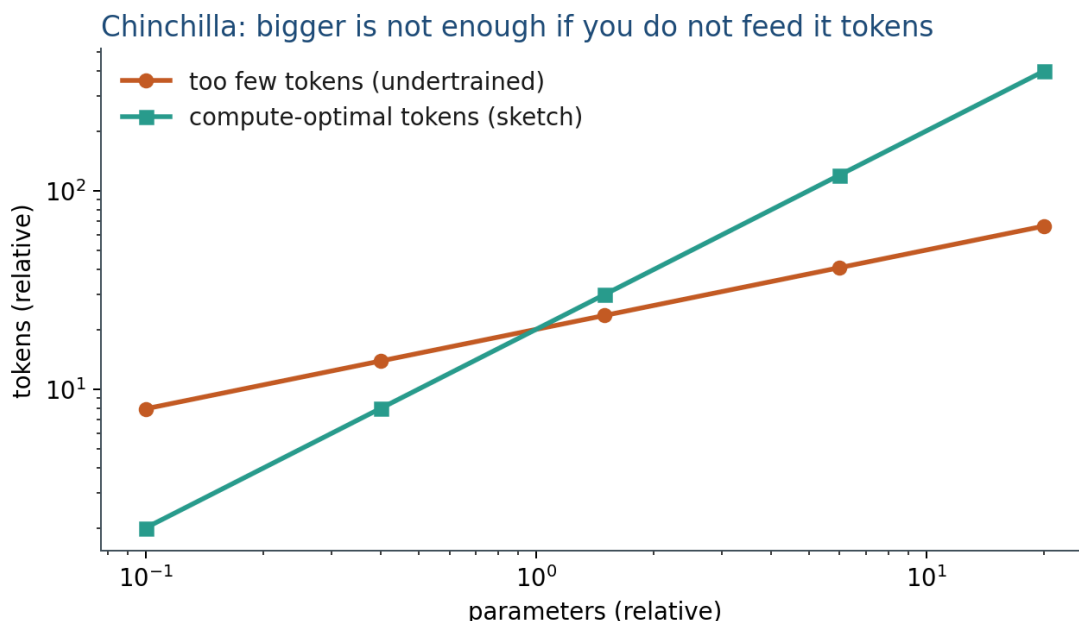


Fix two of $\{N, D, C\}$ and the third is constrained. Early GPT-3-style runs were **compute-optimal for a short training budget** that underused data: a large N , not enough tokens.

Lab 7 will fit $L = aC^{-b} + c$ on constructed points so you see the algebra, not a GPU cluster.

2. Chinchilla: tokens versus parameters

Hoffmann et al. (2022), **Chinchilla**: for a fixed FLOP budget, you should scale **tokens and parameters together**, not dump the budget into a huge model that sees the data once. Their fit said, roughly, equal scaling and about **20 tokens per parameter** for the models they trained.



A smaller model trained longer can beat a larger undertrained one. That is why later LLaMA-style recipes look “small” next to GPT-3 and still read well: they were fed more tokens.

A 7B model at 20 tokens/parameter wants about $7 \times 10^9 \times 20 = 1.4 \times 10^{11}$ tokens. If you only have 14B tokens, you are $10 \times$ short of that rule of thumb—not “we scaled.”

Kaplan-style runs often **oversize** N for a short C : the compute is spent on a wide net that barely sees D . Chinchilla re-spends that C on more tokens. Same FLOP budget, different (N, D) . Write both numbers, not only “7B.”

3. What a law does not tell you

Downstream accuracy is not L . Alignment (Weeks 8–9), data quality, mixture, and context length all sit off the curve. Inference cost scales with N even if you already paid for training. For a project: cite the checkpoint size and the token budget if you know them; do not claim “we scaled” because you trained two extra epochs on a toy set.

4. Teaching this note

About **35 minutes** at the board: write $L(C)$, mark L_∞ , then a Chinchilla arithmetic example (tokens vs parameters at fixed FLOPs). Play Karpathy **after 20:00**, especially **25:43–27:43** (scaling laws, ~2 min,

plus a little context). The fit in Lab 7 is later in the week.

Put 20 tokens/parameter on the board as a **rule of thumb from one paper**, not a law of nature. Then do the $7B \times 20$ arithmetic so “we trained a 7B” is never a complete sentence.

5. Worked example

Suppose $L = 2C^{-0.15} + 1.4$. At $C = 10^4$,

$$L \approx 2 \cdot 10^{-0.6} + 1.4 \approx 2 \cdot 0.251 + 1.4 \approx 1.90.$$

At $C = 10^6$, $10^{-0.9} \approx 0.126$, so $L \approx 2 \cdot 0.126 + 1.4 = 1.65$. A $100\times$ compute jump bought 0.25 nats, not a new architecture. If your measured loss is already 1.42, you are sitting on L_∞ : extra FLOPs buy almost nothing.

Chinchilla, same FLOPs. Training FLOPs scale roughly as $6ND$ (forward+backward). Model A: $N = 4N_0$, $D = D_0 / 4$. Model B: $N = N_0$, $D = D_0$. Same ND . Chinchilla picks the **balanced** pair, not the $4\times$ -larger starved model. If the rule of thumb is 20 tokens/parameter, B matches it when $D_0 = 20N_0$; A has only 5 tokens/parameter.

6. Where students get stuck

- Reading L as “chat quality.” It is pretraining cross-entropy.
 - Thinking a bigger N always wins at fixed FLOPs. Chinchilla is the counterexample.
 - Fitting a, b, c on four noisy toy points and claiming a law. Lab 7 is the algebra; the paper’s range is many orders of magnitude.
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7. Video

[Karpathy — Intro to Large Language Models](#). Play from **20:00**, and pause on **25:43–27:43** (scaling laws). The first 20 min were Week 1; this hour is the compute curve and why tokens matter as much as parameters.

8. Practice

1. Two models, same FLOPs. One has $4\times$ the parameters and $1/4$ the tokens of the other. Who does Chinchilla pick, and why?
2. If L_∞ is already almost your measured loss, what does extra compute buy?
3. Why can a scaling law look great while a chat eval is flat?
4. Using $L = 2C^{-0.15} + 1.4$, compute L at $C = 10^5$. ($10^{-0.75} \approx 0.178$.)

5. A 1B-parameter model trained on 4B tokens. Using 20 tokens/parameter, is it over- or under-trained on data? By what factor?